

Statistics Support Material

Table 1: Glossary

Term	Definition
Association	Association is a measurable, statistical dependence of relationship between two variables, where changes in one variable correspond to changes in another.
Contingency table (<i>crosstab or two-way table</i>)	A contingency table is a type of table that displays the frequencies between two categorical variables.
Correlation	Correlation is a type of statistical measure that describes the size and direction between two variables that are linearly related.
Dimension	In higher statistics, dimension refers to the number of features or variables in a dataset, where more dimensions increase computational complexity
Eigenvalue	An eigenvalue is a scalar value derived from a square matrix that determines the magnitude of stretching or compression along the direction defined by the corresponding eigenvector. <i>See definition of eigenvector.</i> In Principal Component Analysis (PCA), eigenvalues measure how much of the total variance each principal component explains. Larger eigenvalues correspond to components that capture more variability in the data.
Eigenvector	An eigenvector is a non-zero vector that represents the principal direction of a linear transformation. When the transformation is applied, the vector keeps the same direction, changing only in magnitude. It satisfies the equation: $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ where A is the square matrix (linear transformation), v is the eigenvector and λ is the eigenvalue. In Principal Component Analysis (PCA), an eigenvector is a unit vector pointing in the direction of the greatest remaining variance in the data, defining each principal component.
Feature	A feature is an input variable used in modelling to predict or understand a target variable.
Imputation	Imputation refers to the process of replacing missing or inconsistent data points with substituted, plausible values to create a complete dataset for analysis.
Normality (<i>normal or Gaussian distribution</i>)	Normality refers to the assumption that a dataset or distribution follows a normal distribution, characterised by a symmetric, bell-shaped curve where the mean, median and mode are equal.
Standardisation (<i>Z-score normalisation</i>)	Standardisation is the process of transforming data to a common scale, typically centres around a mean of zero and scaling to a standard deviation of 1.
Target variable	A target variable is the primary outcome, response or dependent variable that a model aims to predict and explain using features.
Variable	A variable is any measurable characteristic, attribute, number or quantity that varies among individuals, objects or over time, and are fundamental components of datasets and often represented as columns.
Variance	Variance measures how spread out a set of numbers is from its average (mean), calculated as the average of the squared differences from the mean.

Figure 1: Types of data and scales of measurement

Source: <https://studyonline.unsw.edu.au/blog/types-of-data>

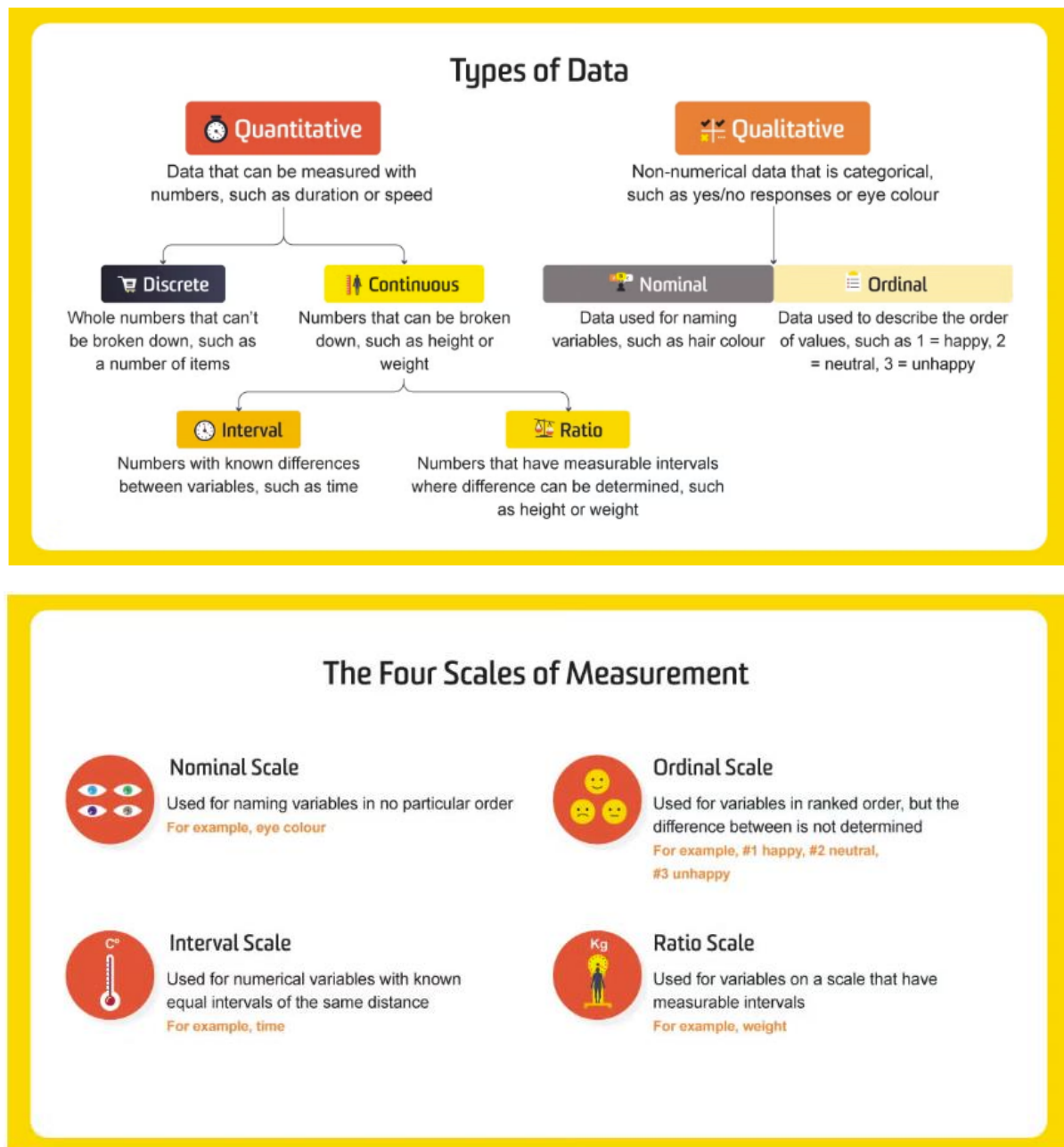


Table 3: Missing values

Method	What It Is	Assumptions	Pros	Cons
Listwise Deletion	Remove rows with missing values	Missing At Random (MAR); sample size remains adequate	Simple; no modelling assumptions	Data loss; biased if missingness not random
Mean/Median/Mode Imputation	Replace missing values with central tendency	MAR; central value is representative	Easy; good for small missingness	Reduces variance; distorts relationships
Ordinal Imputation	Replace with median category or proportional draw	Ordinal scale meaningful; missingness not extreme	Respects ordering; simple	Reduces variability
Hot-Deck Imputation	Replace with values from similar respondents	“Similar” cases identifiable; missingness relates to observed traits	Preserves realistic values; maintains variability	More complex; requires similarity definition
Regression Imputation	Predict missing values using other variables	Stable relationships; MAR; correct model specification	More informed than simple imputation	Inflates correlations; adds modelling assumptions
Treat Missingness as a Category	Add “Unknown” as a category	Missingness may be meaningful	No data loss; sometimes predictive	Not suitable for ordinal variables

Table 4: Correlations & associations

Method	Use When	Assumptions	Strengths	Limitations
Pearson Correlation	Numeric, continuous variables	Linearity; normality; no major outliers	Simple; widely used	Not suitable for ordinal or non-normal data
Spearman's Rho	Ordinal or non-normal numeric data	Monotonic relationships; rankable data	Robust; works with survey data; easy to interpret	Doesn't model latent traits
Kendall's Tau	Ordinal data with many ties or small samples	Monotonic relationships	Robust to ties; good for small N	More conservative; smaller effect sizes
Chi-Square Test	Categorical–categorical	Adequate expected cell counts; independence	Good for category associations	Not for ordinal or numeric data
ANOVA	Numeric trait across groups	Normality; equal variances	Tests group differences	Sensitive to assumption violations
Kruskal-Wallis	Ordinal or non-normal numeric data	Independent groups	Non-parametric alternative to ANOVA	One variable at a time
Mutual Information	Detecting nonlinear relationships	Variables can be discretised	Captures nonlinear patterns	Harder to interpret
Polychoric Correlation	Ordinal–ordinal latent traits	Underlying continuous latent variables; thresholds; normality	Ideal for psychometrics	Heavy computation; fragile; limited to ordinal pairs
Tetrachoric Correlation	Binary–binary latent traits	Same latent assumptions as polychoric	Good for binary latent traits	Only works for binary variables; computationally heavy

Table 5: Correlations thresholds for relationship strength

NB: The below table provided general guidance. Thresholds are subjective and highly dependent on the field of study.

Relationship Strength	Lower (inclusive)	Upper (exclusive)
None or negligible relationship	0	0.1
Weak	0.1	0.4
Moderate	0.4	0.7
Strong	0.7	0.9
Very strong to perfect	0.9	1

Table 6: Assumptions of Principal Component Analysis (PCA)

Assumption	Description
Linearity	PCA assumes that the main patterns in the data can be captured through linear combinations of variables.
Variables should reflect meaningful relationships	PCA assumes real structure in the data. If relationships are weak, random, or unstable, PCA will not extract meaningful components.
Comparable variable scales	PCA is sensitive to scale; variables should be standardized if they have different units or ranges.
Orthogonal components	PCA assumes principal components are uncorrelated and orthogonal, which may not reflect true underlying structure.
No significant outliers	Outliers can distort variance and rotate principal components, so PCA assumes data is reasonably clean.
Numeric variables	PCA requires numeric variables.

Table 7: Diagnostics for Principal Component Analysis (PCA)

Diagnostic	Description
Kaiser-Meyer-Olkin (KMO) test	Evaluates whether correlations are compact and patterned enough for PCA to extract structure.
Bartlett's test of sphericity	Tests whether the correlation matrix differs significantly from an identity matrix.
Inspection of the correlation matrix	PCA requires variables to have non-trivial correlations.
Assessment of outliers and distribution shape	Outliers distort covariance and correlation matrices, whilst skewed or heavy-tailed variables may require transformation.
Adequate sample size	PCA does not require a minimum sample size mathematically, but stable correlation estimates typically require at least 5–10 observations per variable.

Table 8: Kaiser-Meyer-Olkin (KMO) test and Bartlett's test of sphericity

The **Kaiser-Meyer-Olkin (KMO) test** assesses how suitable a dataset is for factor analysis by measuring sampling adequacy for each observed variable in the model as well as the complete model, using correlation and partial correlation. KMO varies from 0 to 1.0 and KMO overall should be 0.60 or higher to proceed.

KMO test interpretation	Lower (inclusive)	Upper (exclusive)
Unacceptable	0	0.5
Terrible	0.5	0.6
Mediocre	0.6	0.7
Average	0.7	0.8
Meritorious	0.8	0.9
Marvelous	0.9	1

Reference: Kaiser, Henry F. 1974. "An Index of Factorial Simplicity." Psychometrika 39 (1): 31–36.

Bartlett's test of sphericity is a statistical test used to determine whether the correlation matrix contains significant relationships among variables. It evaluates the null hypothesis that the correlation matrix is an identity matrix, meaning all correlations are zero. A significant result ($p < 0.05$) indicates that the variables share enough structure for PCA or factor analysis to be appropriate.